

Project Report

1. INTRODUCTION

1.1 Project Overview

The **Cosmetic Insights** project is an interactive business intelligence dashboard developed using **Tableau Public** to analyze cosmetic product data. The dashboard is designed to transform raw product information into meaningful visual insights, enabling users to better understand product performance across different brands, categories, prices, rankings, and skin types.

The project uses the **Cosmetics.xlsx** dataset, which contains approximately **1,472 product records and 11 attributes**, including product name, brand, category, price, rank, and skin-type suitability. Before visualization, the dataset was verified for data quality, and a calculated field named **All Skin Types** was created to identify products suitable for multiple skin types.

The dashboard consists of **11 interactive visualizations, 3 KPI cards, and 8 story points**, allowing users to explore cosmetic products through dynamic filters and graphical representations. Visualizations such as packed bubble charts, bar charts, pie charts, scatter plots, and KPI indicators help identify top-performing brands, category-wise product distribution, pricing patterns, product rankings, and skin-type suitability.

Overall, the **Cosmetic Insights** dashboard provides an intuitive and interactive platform for product managers, business analysts, retailers, and marketing professionals to analyze cosmetic product data efficiently and support data-driven business decision-making.

1.2 Purpose

The primary purpose of the **Cosmetic Insights** project is to develop an interactive Tableau dashboard that provides meaningful insights into cosmetic product data through effective data visualization. The dashboard helps users analyze product information based on brands, categories, prices, rankings, and skin-type suitability in a simple and interactive manner.

The project aims to reduce the effort required to analyze large volumes of cosmetic product data by presenting key performance indicators (KPIs) and visualizations in a single dashboard. Interactive filters enable users to explore specific brands and categories, making the analysis more flexible and user-friendly.

The dashboard also supports product managers, business analysts, retailers, and marketing professionals in identifying top-performing brands, understanding product

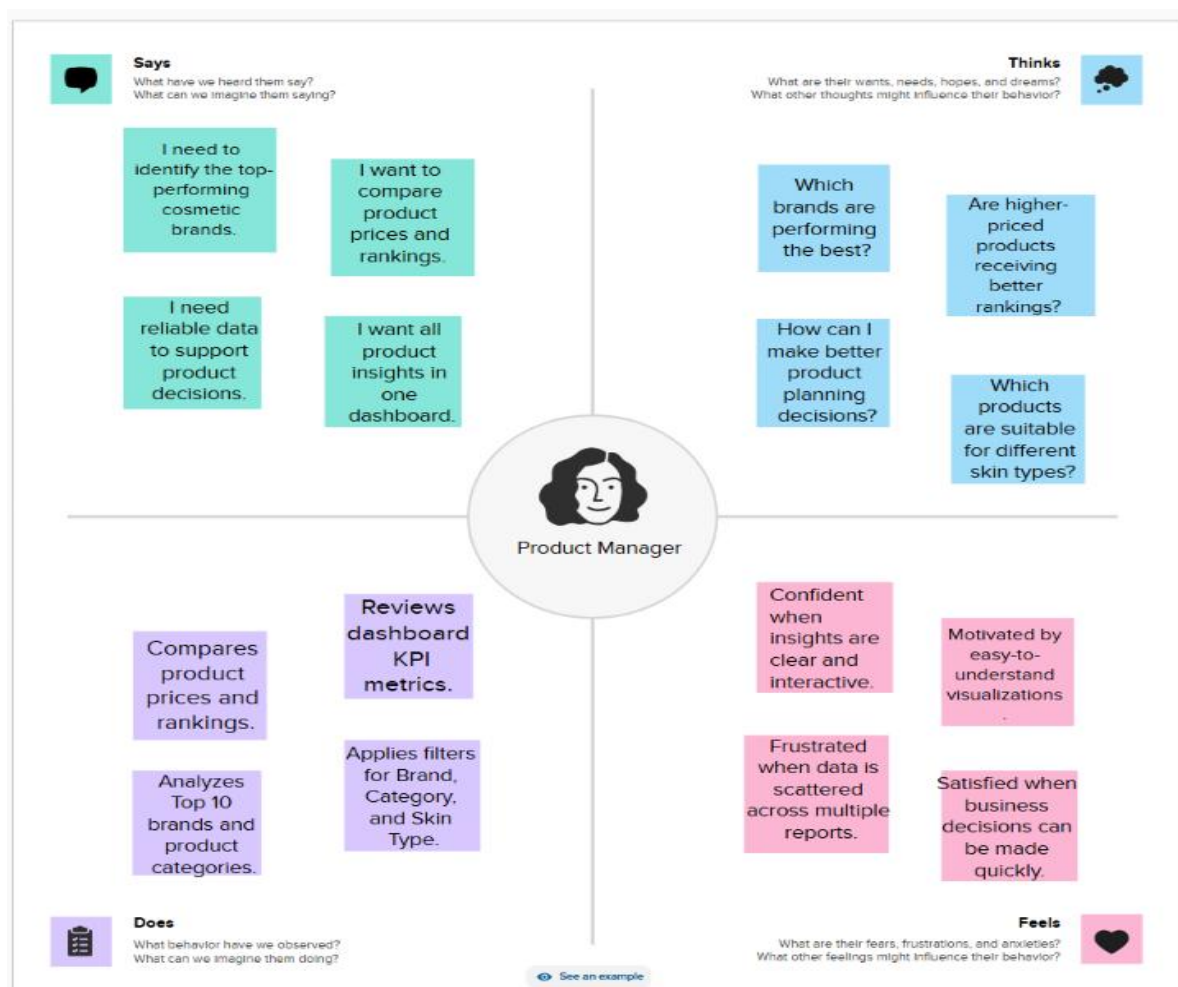
distribution across categories, comparing product prices and rankings, and analyzing products suitable for different skin types. These insights help organizations make informed, data-driven decisions related to product planning, marketing strategies, and business growth.

2. IDEATION PHASE

2.1 Problem Statement

The increasing volume of cosmetic product data makes it difficult to manually analyze product performance, compare brands, categories, prices, rankings, and skin-type suitability. The **Cosmetic Insights** dashboard addresses this problem by providing interactive visualizations and KPI indicators that enable efficient data analysis and support informed business decision-making.

2.2 Empathy Map Canvas



2.3 Brainstorming

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP

You can select a sticky note and hit the pencil [switch to sketch] icon to start drawing!

Person 1



3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

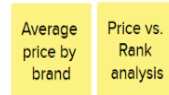
TIP

Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mural.

Group 1 – Product Analysis



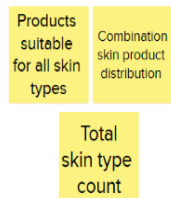
Group 2 – Price Analysis



Group 4 – Dashboard & Reporting



Group 3 – Skin Type Analysis



4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes

TIP

Participants can use their cursors to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the laser pointer holding the **H** key on the keyboard.



5

After you collaborate

You can export the mural as an image or pdf to share with members of your company who might find it helpful.

Quick add-ons

- Share the mural**
Share a view link to the mural with stakeholders to keep them in the loop about the outcomes of the session.
- Export the mural**
Export a copy of the mural as a PNG or PDF to attach to emails, include in slides, or save in your drive.

Keep moving forward

- Strategy blueprint**
Define the components of a new idea or strategy.
[Open the template →](#)
- Customer experience journey map**
Understand customer needs, motivations, and obstacles for an experience.
[Open the template →](#)
- Strengths, weaknesses, opportunities & threats**
Identify strengths, weaknesses, opportunities, and threats (SWOT) to develop a plan.
[Open the template →](#)

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Scenario: [Existing experience through a product or service]	 Entice How does someone become aware of this service?	 Enter What do people experience as they begin the process?	 Engage In the core moments in the process, what happens?	 Exit What do people typically experience as the process finishes?	 Extend What happens after the experience is over?
 Experience steps What does the person (or people) at the center of this scenario typically experience in each step?	Receives the dashboard link and learns about the dashboard.	Opens the dashboard and views KPI cards and visualizations.	Uses filters, explores charts, and analyzes product information.	Completes the analysis and reviews the dashboard insights.	Revisits or shares the dashboard for future analysis.
 Interactions What interactions do they have at each step along the way? • People: Who do they see or talk to? • Places: Where are they? • Things: What digital touchpoints or physical objects do they use?	Opens the Tableau Public dashboard link. Opens the Tableau Public dashboard link.	Dashboard interface, KPI cards, charts.	Brand filter, Category filter, Interactive charts.	Dashboard story, summary insights.	Tableau Public sharing options.
 Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Access cosmetic product insights quickly.	Get an overview of cosmetic products and brands.	Compare brands, categories, prices, rankings, and skin types.	Make informed business decisions based on the analysis.	Monitor product trends and share insights with others.
 Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Easy online access.	Clean and interactive dashboard layout.	Interactive filtering and dynamic visualizations.	Easy interpretation of business insights.	Dashboard is available online anytime.
 Negative moments What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?	User may not know the dashboard features initially.	Too much information may appear at first glance.	Users may require time to understand some charts.	No direct export option in Tableau Public.	Requires an internet connection.
 Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Add a dashboard introduction or user guide.	Highlight key insights or provide navigation tips.	Add tooltips and chart descriptions.	Enable export functionality in future versions.	Add more datasets and advanced analytics in future versions. See an example

3.2 Solution Requirements

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Import Cosmetic Dataset	Load cosmetic dataset into Tableau. Validate and clean the imported data before analysis.
FR-2	Display KPI Metrics	Show Total Products, Total Brands, Average Price, and Skin-Type Count.

FR-3	Generate Interactive Dashboard	Display charts for brand analysis, category analysis, pricing, and skin-type analysis for cosmetic product analysis.
FR-4	Apply Interactive Filters	Allow filtering by Brand, Category, and Skin Type for customized analysis.
FR-5	Perform Product Analysis	Display Top 10 Brands, Product Distribution, and Price vs. Rank analysis to identify business insights.
FR-6	Support Data Analysis	Enable comparison of brands, categories, prices, and skin-type suitability through interactive visualizations.

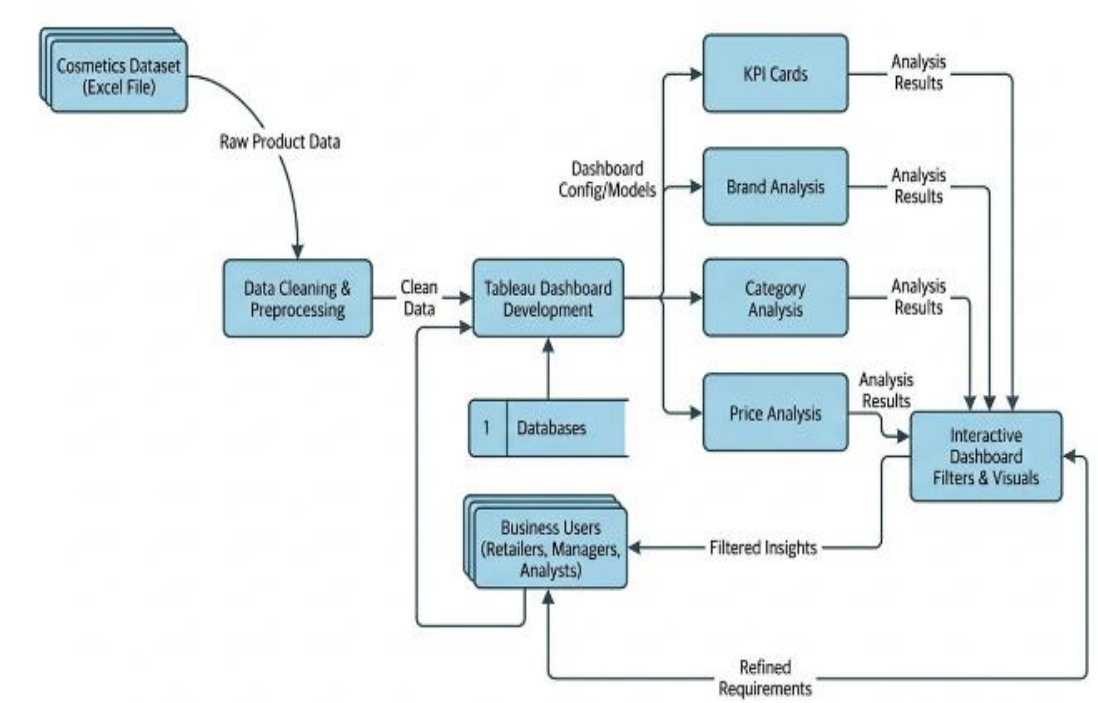
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard provides a simple, interactive, and easy-to-understand interface so users can navigate visualizations and filters without difficulty.
NFR-2	Security	Only authorized users would be able to access or modify the Tableau dashboard and its underlying dataset.
NFR-3	Reliability	The dashboard would consistently provide accurate, reliable, and up-to-date visualizations based on the available cosmetic dataset.
NFR-4	Performance	Dashboard visualizations and filters would load and respond quickly to provide a smooth user experience.
NFR-5	Availability	The dashboard would be accessible whenever users need to analyze cosmetic product data using Tableau Desktop, Tableau Public, or Tableau Cloud.
NFR-6	Scalability	The solution would support larger datasets, additional cosmetic brands, new visualizations, and future enhancements without significant redesign.

3.3 Data Flow Diagram

DFD Level-0 (Context Diagram)



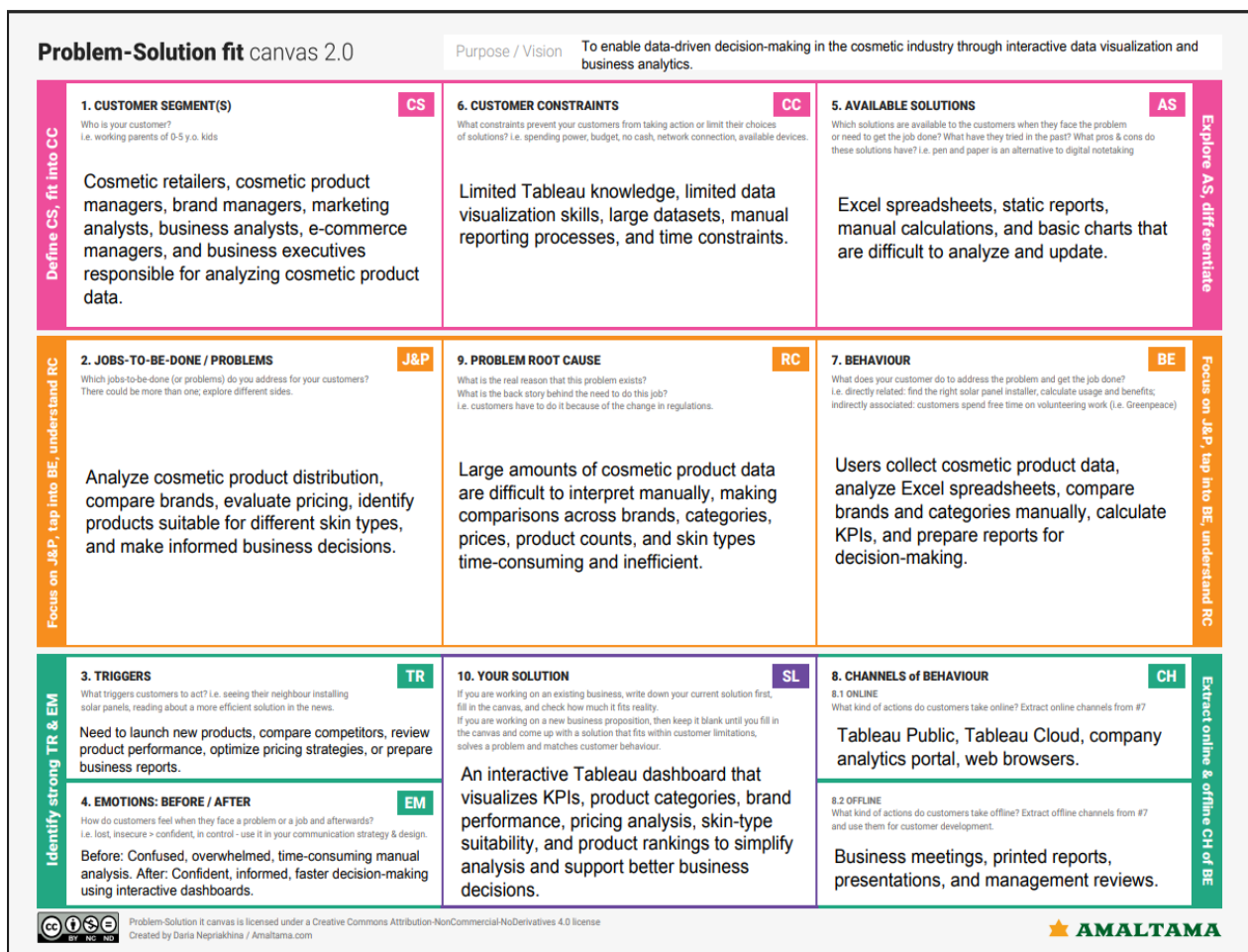
3.4 Technology Stack

Component	Description	Technology Used
Dataset	Cosmetic product dataset used for analysis	Microsoft Excel (.xlsx)
Data Source	Source of data imported into Tableau	Cosmetics.xlsx
Data Visualization Tool	Creating charts, KPI cards, dashboards, and stories	Tableau Public
Dashboard Development	Designing interactive dashboards and stories	Tableau Public Desktop Application
Calculated Fields	Creating custom metrics for analysis	Tableau Calculated Fields
Interactive Features	Dynamic filtering and user interaction	Tableau Filters & Dashboard Actions
Dashboard Publishing	Publishing and sharing the dashboard online	Tableau Public Cloud

File Storage	Storing the dataset and project files	Local File System
Documentation	Preparing the project report and documentation	Microsoft Word
Version Control	Managing project documentation and files	GitHub

4. PROJECT DESIGN

4.1 Problem Solution Fit

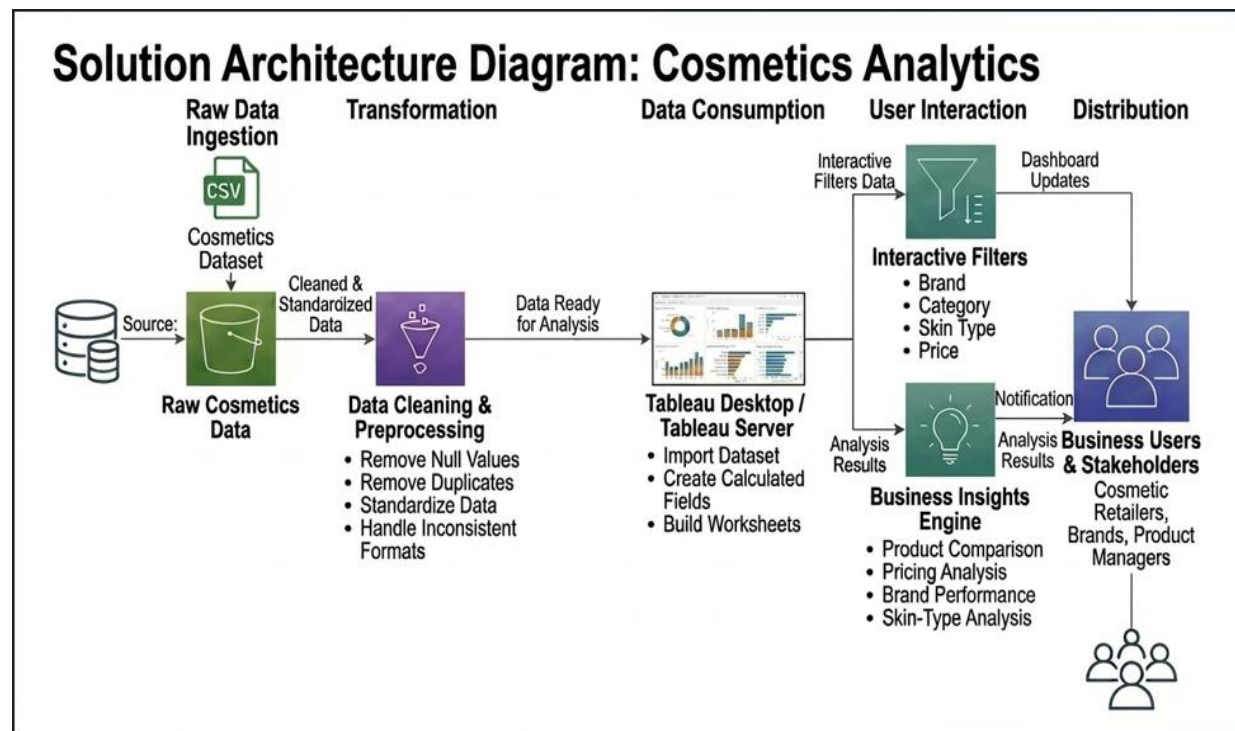


4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Cosmetic businesses often struggle to analyze large volumes of product data across multiple brands, categories, prices, and skin types. Manual analysis using spreadsheets is time-consuming, error-prone,

		and makes it difficult to identify trends and support data-driven decision-making.
2.	Idea / Solution description	Develop an interactive Tableau dashboard that visualizes cosmetic product data through KPI cards, bar charts, pie charts, packed bubble charts, and scatter plots. The dashboard enables users to analyze product categories, brand performance, pricing patterns, skin-type suitability, and product rankings using interactive filters and visualizations.
3.	Novelty / Uniqueness	The dashboard consolidates multiple business insights into a single interactive platform. It provides dynamic filtering, calculated fields (such as All Skin Types), Top-10 brand analysis, and multiple visualizations that simplify product analysis and support faster, data-driven decisions.
4.	Social Impact / Customer Satisfaction	The solution helps retailers and business professionals analyze cosmetic products more efficiently, reducing manual effort and improving decision-making. It enables quicker identification of product trends, pricing patterns, and skin-type suitability, leading to better inventory planning and improved customer satisfaction through informed business strategies.
5.	Business Model (Revenue Model)	The dashboard can be offered as a Business Intelligence (BI) analytics solution through subscription-based Tableau Cloud services, enterprise dashboard deployment, consulting services, or customized analytics solutions for cosmetic retailers, brands, and e-commerce companies.
6.	Scalability of the Solution	The dashboard can be easily scaled by integrating larger datasets, real-time product information, customer reviews, sales data, inventory management, and additional cosmetic brands. It can also be adapted for other retail industries such as skincare, healthcare, fashion, and consumer products with minimal modifications.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect the Cosmetics.xlsx dataset for analysis.	2	High	Rohini Maddineni
Sprint-1		USN-2	Import the dataset into Tableau Public.	1	High	Rohini Maddineni
Sprint-1	Data Preparation	USN-3	Verify data quality by checking missing values and duplicates.	2	High	Rohini Maddineni
Sprint-1		USN-4	Create the calculated field All Skin Types .	3	High	Rohini Maddineni
Sprint-1		USN-5	Validate data consistency and prepare it for visualization.	2	Medium	Rohini Maddineni
Sprint-2	Data Visualization	USN-6	Create charts to analyze brands, categories, prices, and rankings.	3	High	Rohini Maddineni
Sprint-2	Dashboard Development	USN-7	Design KPI cards and interactive dashboard.	5	High	Rohini Maddineni
Sprint-2		USN-8	Add interactive filters for Brand and Category.	2	Medium	Rohini Maddineni
Sprint-2	Story Development	USN-9	Create Tableau Story using dashboard insights.	3	Medium	Rohini Maddineni
Sprint-2	Dashboard Publishing	USN-10	Publish the dashboard on Tableau Public.	2	Medium	Rohini Maddineni
Sprint-2	Documentation	USN-11	Prepare project documentation and final report.	3	High	Rohini Maddineni

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Functional testing was performed to verify that all dashboard components and interactive features work correctly and produce the expected results. Each visualization, KPI card, filter, and story point was tested to ensure accurate data representation and smooth user interaction.

Functional Testing Table

Test Case ID	Feature Tested	Expected Result	Actual Result	Status
FT-01	Dataset Import	Cosmetics.xlsx dataset is imported successfully into Tableau.	Dataset imported successfully.	Pass
FT-02	KPI Cards	KPI cards display correct values for Total Cosmetics, Total Brands, and Total Skin Types.	Values displayed correctly.	Pass
FT-03	Brand Filter	Dashboard updates according to the selected brand.	Filter works correctly.	Pass
FT-04	Category Filter	Visualizations update based on the selected category.	Filter works correctly.	Pass
FT-05	Calculated Field	"All Skin Types" calculated field returns correct results.	Calculated values are accurate.	Pass
FT-06	Visualizations	Charts display accurate product analysis.	All charts displayed correctly.	Pass
FT-07	Dashboard	Dashboard loads with all visualizations and interactive components.	Dashboard functions correctly.	Pass
FT-08	Story	Story navigation displays all story points in sequence.	Story works correctly.	Pass
FT-09	Dashboard Publishing	Dashboard is accessible through Tableau Public.	Dashboard published successfully.	Pass

6.2 Performance Testing

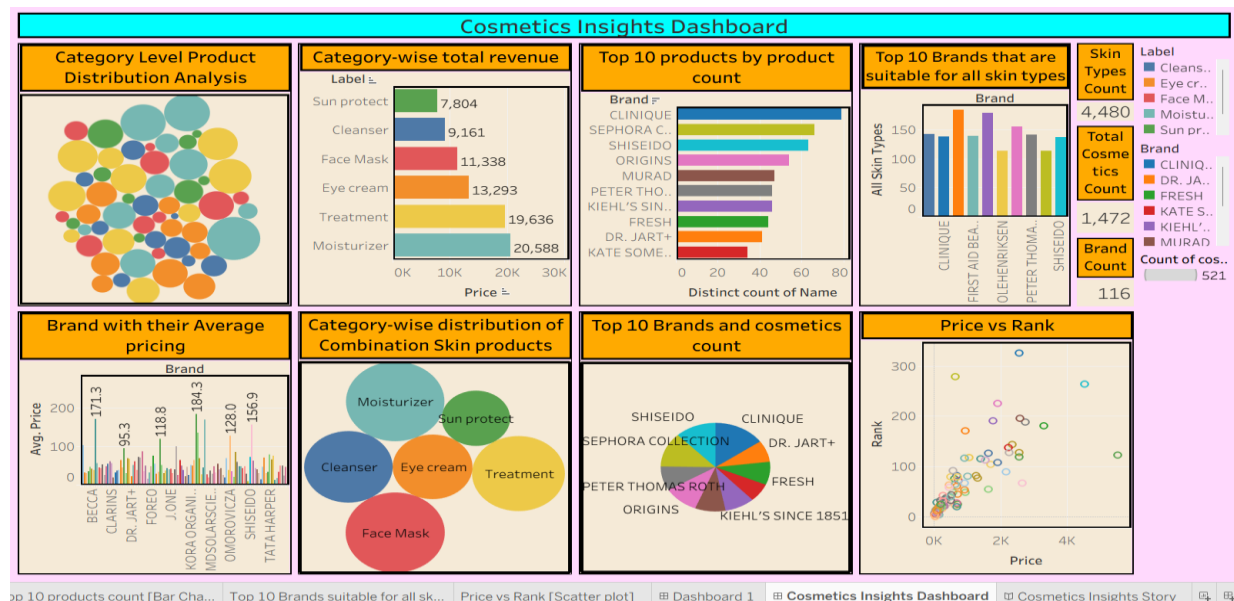
Performance testing was conducted to evaluate the responsiveness, usability, and overall performance of the Tableau dashboard. The dashboard was tested to ensure that it loads quickly, responds efficiently to user interactions, and provides accurate visualizations without noticeable delays.

Performance Testing Table

Parameter	Expected Result	Observed Result	Status
Dashboard Loading Time	Dashboard loads within a few seconds.	Loaded successfully within acceptable time.	Pass
Filter Response Time	Filters update visualizations immediately.	Filters responded quickly.	Pass
Visualization Rendering	Charts render without errors or delays.	All visualizations displayed correctly.	Pass
Story Navigation	Story transitions smoothly between story points.	Navigation was smooth.	Pass
Dashboard Availability	Dashboard is accessible online through Tableau Public.	Dashboard available successfully.	Pass
Data Accuracy	Dashboard displays accurate information from the dataset.	Results matched the dataset.	Pass
User Interaction	Users can interact with filters and visualizations without issues.	Interactions worked smoothly.	Pass

7. RESULTS

7.1 Output Screenshots



8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboard.
- Easy product analysis.
- Dynamic filters.
- Better business decisions.
- Easy data visualization.
- Quick identification of top-performing brands.

Disadvantages

- Dataset is static.
 - Limited to available product information.
 - No real-time data updates.
 - Tableau Public requires internet for online access.
 - Dashboard depends on data quality.
-

9. CONCLUSION

The **Cosmetic Insights** project successfully developed an interactive Tableau dashboard for analyzing cosmetic product data. By using effective data visualizations, KPI cards, interactive filters, and story points, the dashboard transforms raw product data into meaningful business insights. It enables users to analyze cosmetic products based on brands, categories, prices, rankings, and skin-type suitability in a clear and interactive manner.

The dashboard simplifies the process of exploring large volumes of cosmetic product information and supports product managers, business analysts, retailers, and marketing professionals in making informed, data-driven decisions. Features such as dynamic filtering, calculated fields, and interactive visualizations enhance the overall user experience and improve the efficiency of data analysis.

Overall, the project demonstrates how business intelligence tools like **Tableau Public** can effectively convert raw data into actionable insights, helping organizations better understand product trends and support strategic decision-making in the cosmetics industry.

10. FUTURE SCOPE

The **Cosmetic Insights** dashboard can be further enhanced by incorporating additional features and advanced analytics to provide deeper business insights. Some of the possible future improvements are:

- Integrate **real-time cosmetic product data** from online retail platforms to enable live analysis and up-to-date reporting.
- Include **customer ratings, reviews, and sales data** to better understand product performance and customer preferences.
- Apply **machine learning techniques** to predict product trends, recommend products, and forecast future demand.
- Expand the dashboard by adding **inventory, sales, and customer analytics** to provide a more comprehensive business intelligence solution.
- Deploy the dashboard on **Tableau Cloud or Tableau Server** to support secure access, collaboration, and organization-wide reporting.
- Develop a **mobile-friendly dashboard** with additional interactive features to improve accessibility and user experience across different devices.

These enhancements will increase the usefulness of the dashboard and help businesses make more accurate, data-driven decisions in the cosmetics industry.

11. APPENDIX

Dataset link:

<https://1drv.ms/x/c/990d42432ebec19d/IQADEOrQqg3nR6ACnJuGkTQIAZcPSj0BAuZfZ4eKzLhe1sA?e=1C52rV>

Dashboard link:

https://public.tableau.com/app/profile/rohini.maddineni/viz/CosmeticsInsightsDashboard_17828076629250/CosmeticsInsightsDashboard

Story link:

https://public.tableau.com/app/profile/rohini.maddineni/viz/CosmeticsInsightsStory_17828108328690/CosmeticsInsightsStory

Github link:

<https://github.com/mrohini2911/Cosmetic-Insights>

Project demo link:

<https://drive.google.com/file/d/1pFPu0tkLaMuL-dsKF4IF2GEEtRKo8FQ8/view?usp=sharing>